

Monte Carlo Estimation of Option Delta

Oriol Martínez Sunyer

1 Introduction

Understanding how the value of a financial derivative varies as a function of the price of its underlying asset is of great relevance in areas of finance such as derivatives trading, hedging and risk management. The Delta of a financial derivative measures precisely this dependence: it is defined as the partial derivative of the derivative's price with respect to the price of the underlying asset. For many options, however, it is difficult to obtain exact values of this measure or closed-form expressions for it, even under a specified mathematical model.

This project implements and studies Monte Carlo estimators for the Delta of European, Asian, digital and barrier call options under the assumptions of the Black-Scholes-Merton (BSM) model. In particular, it studies finite-difference methods, pathwise differentiation and the likelihood-ratio method, comparing their performance across the different options and a range of parameter values. Mean squared error (MSE) and runtime are used as the primary measures of estimator performance. The project additionally implements variance-reduction techniques to improve estimation accuracy and considers modifications of the standard methods that extend their use to payoffs for which the naive estimators are unsuitable. Although closed-form Delta expressions exist for most of the specific contracts considered, these contracts are useful both as benchmarks and as representatives of distinct payoff classes, including terminal, multi-step, discontinuous and continuously monitored payoffs. The broader aim is to determine how estimator suitability varies with payoff structure, so that the conclusions may guide estimator choice for similar options for which no closed-form BSM Delta expression is available.

2 Background

We begin by outlining the principal definitions, model and estimators used throughout this investigation. This section is intended only to establish the notation and results required to interpret the numerical experiments. Precise derivations of the estimators presented here can be found in the accompanying `notebooks/01_estimator_theory.ipynb` file.

2.1 Payoff Definitions

The payoff functions of the derivatives considered are given in Table 1 below.

Table 1: Payoff functions of the derivatives considered in the project.

Derivative	Payoff H
European call	$(S_T - K)^+$
Arithmetic Asian call	$\left(\frac{1}{M+1} \sum_{m=0}^M S_{t_m} - K\right)^+$
Digital call	$Q \mathbf{1}_{\{S_T > K\}}$
Up-and-out barrier call, discrete monitoring	$(S_T - K)^+ \mathbf{1}_{\{\max_{0 \leq m \leq M} S_{t_m} < B\}}$

Here, S_t denotes the price of the underlying asset at time t , K is the strike price, T is the maturity, M is the number of time intervals for the Asian call and the number of monitoring intervals for the barrier call, Q is the fixed cash payment of the digital call, and B is the upper barrier. The notation $(x)^+ = \max(x, 0)$, while $\mathbf{1}_A$ denotes the indicator of an event A . For the Asian and barrier calls, $0 = t_0 < t_1 < \dots < t_M = T$ are the averaging/monitoring dates.

The barrier case requires one additional distinction. Although the asset paths are necessarily simulated on a discrete time grid, the reference Delta used throughout the barrier investigation corresponds to the continuously monitored option, whose payoff is defined by

$$H_{\text{UO}}^{(c)} = (S_T - K)^+ \mathbf{1}_{\{\sup_{0 \leq t \leq T} S_t < B\}}.$$

Here, the superscript c denotes continuous monitoring. Brownian-bridge conditioning is used in the adapted estimators to account for possible barrier crossings between consecutive simulated values. The continuously monitored contract is considered because its Delta under the Black-Scholes-Merton model is available in closed form, providing a reference value for the calculation of the MSE.

2.2 Mathematical Model

The simulations and option-specific forms of the estimators used throughout this project are built under the assumptions of the Black-Scholes-Merton (BSM) model. Under the risk-neutral measure, the underlying asset price follows a geometric Brownian motion with constant risk-free rate and volatility. It satisfies the SDE and has terminal solution

$$\begin{aligned} dS_t &= rS_t dt + \sigma S_t dW_t, \\ S_T &= S_0 \exp \left(\left(r - \frac{1}{2}\sigma^2 \right) T + \sigma\sqrt{T} Z \right), \quad Z \sim N(0, 1). \end{aligned}$$

Here, S_0 is the initial asset price, r is the continuously compounded risk-free interest rate, σ is the volatility of the underlying asset, W_t is a standard Brownian motion under the risk-neutral measure, T is the time to maturity, and $Z = W_T/\sqrt{T}$ is a standard normal random variable (Black and Scholes, 1973; Merton, 1973).

For an option with terminal payoff H , risk-neutral valuation gives

$$V(S_0) = \mathbb{E} [e^{-rT} H].$$

For certain options, this expectation and its Delta can be evaluated analytically under the model. For others, no closed-form expression is available, so a numerical approximation is necessary. In cases where an analytical Delta is available, it is used throughout the investigation as the reference value against which the Monte Carlo estimates are compared.

2.3 Delta Estimators

Let $Y(S_0, Z)$ denote one discounted payoff observation generated from the random input Z . The option value and its standard Monte Carlo estimator may then be written as

$$V(S_0) = \mathbb{E} [Y(S_0, Z)], \quad \hat{V}_N(S_0) = \frac{1}{N} \sum_{i=1}^N Y(S_0, Z_i).$$

Here, $Z_1, \dots, Z_N$ are independent random inputs used to generate N underlying asset paths. Pro-

vided that the relevant expectation exists, the Law of Large Numbers guarantees that $\hat{V}_N(S_0)$ converges to $V(S_0)$ as N increases. We consider three approaches to Monte Carlo Delta estimation, with both forward and central variants of the finite-difference method. Their generic forms are given in Table 2 below.

Table 2: Generic forms of the Monte Carlo Delta estimators considered.

Estimator	Generic form
Forward difference	$\hat{\Delta}_{\text{FD}}(N, h) = \frac{\hat{V}_N(S_0 + h) - \hat{V}_N(S_0)}{h}$
Central difference	$\hat{\Delta}_{\text{CD}}(N, h) = \frac{\hat{V}_N(S_0 + h) - \hat{V}_N(S_0 - h)}{2h}$
Pathwise differentiation	$\hat{\Delta}_{\text{PW}} = \frac{1}{N} \sum_{i=1}^N \frac{\partial Y(S_0, Z_i)}{\partial S_0}$
Likelihood-ratio	$\hat{\Delta}_{\text{LR}} = \frac{1}{N} \sum_{i=1}^N \left[\frac{\partial Y(X_i; S_0)}{\partial S_0} + Y(X_i; S_0) W(X_i) \right]$

Here, $h > 0$ is the finite-difference step size, X_i denotes the simulated terminal value or collection of path values on the i -th path, and $f(x; S_0)$ denotes the density of X_i . The likelihood-ratio score is

$$W(X) = \frac{\partial}{\partial S_0} \log f(X; S_0).$$

The derivative $\partial Y(X_i; S_0)/\partial S_0$ in the likelihood-ratio estimator measures any explicit dependence of the discounted payoff on S_0 while X_i is held fixed; it is zero for the standard payoff representations considered here, but is required for the Brownian-bridge barrier estimator. Finite differences perturb the initial asset price and compare the resulting price estimates. Pathwise differentiation instead differentiates each simulated payoff observation before averaging, while the likelihood-ratio method differentiates the probability density through its score (Broadie and Glasserman, 1996; Glasserman, 2004; Haugh, 2017). The option-specific forms of the pathwise derivative and likelihood-ratio score depend on the payoff and the simulated path representation being considered. Their derivations, together with a more complete explanation of why the methods work, are presented in `notebooks/01_estimator_theory.ipynb`.

3 Implementation

The estimators were implemented entirely in Python, while the numerical analysis was conducted in Jupyter notebooks. Three source files form a small Monte Carlo pricing framework, which is used by three further modules within the `estimators` directory to implement the methods investigated. More specifically, the purpose of each file is as follows:

- `path_simulator.py`: generates terminal asset values and multi-step price paths under the BSM model. Its functions accept predetermined Gaussian inputs, allowing Common Random Numbers (CRN) to be used across related simulations.
- `payoffs.py`: implements the payoff observations for each derivative considered, using the simulated terminal values or paths.
- `mc_pricer.py`: combines the path simulator and payoff functions to produce option-price estimates as discounted sample means.
- `finite_difference.py`: contains the forward- and central-difference Delta estimators for each

option, including their CRN implementations.

- `pathwise.py`: contains the pathwise-differentiation estimators in both their naive and option-specific adapted forms.
- `likelihood_ratio.py`: contains the ordinary and payoff-centred likelihood-ratio estimators, together with their adaptations for continuously monitored barrier options.
- The `.ipynb` files contain the theoretical and numerical analysis of the estimators. They also include local helper functions used to calculate reference values and option-specific tuning parameters required by the experiments.

The core calculations operate on NumPy arrays containing the simulated and processed data. This allows the calculations to be vectorised across asset paths, simplifying the manipulation of the simulations and reducing execution times.

Some estimators require additional modifications for particular payoff structures. The pathwise digital-call estimator uses a smooth approximation to the discontinuous payoff function. For barrier calls, Brownian-bridge conditional survival probabilities account for possible barrier crossings between consecutive simulated values, allowing the estimators to target a continuously monitored contract from a finite path skeleton. These adaptations extend the methods to payoffs for which their naive forms are unsuitable. Their mathematical justification is discussed in `notebooks/01_estimator_theory.ipynb`.

4 Experimental Design

The experiments conducted to assess the appropriateness of each of the Delta-estimation methods across options focus on two primary metrics: mean squared error (MSE), in either its standard or relative form, and Runtime $\times$ MSE. For R independent trials, the estimated MSE is defined as

$$\widehat{\text{MSE}} = \frac{1}{R} \sum_{r=1}^R \left(\widehat{\Delta}^{(r)} - \Delta_{\text{ref}} \right)^2,$$

where $\widehat{\Delta}^{(r)}$ is the estimate obtained in trial r and Δ_{ref} is the corresponding reference Delta. Relative MSE is defined as

$$\widehat{\text{rMSE}} = \frac{\widehat{\text{MSE}}}{\Delta_{\text{ref}}^2}.$$

It is useful when the magnitude of Delta varies considerably across the parameter grid, although it is not interpreted in regions where the reference Delta is zero or very close to zero.

The runtime of each estimator was averaged across the same repeated trials. The metric Runtime $\times$ MSE combines accuracy and computational cost, penalising methods that require considerably more time to achieve a given level of accuracy. These metrics therefore allow both the numerical accuracy and practical efficiency of the estimators to be considered.

To compare the numerical Delta estimates against an underlying reference value, known closed-form BSM formulas were used for the European, digital and continuously monitored barrier calls. For the arithmetic Asian call, for which no corresponding closed-form Delta is available, an independent pathwise-differentiation estimate using 10^7 simulations was used as a high-precision reference. This value is itself a Monte Carlo estimate rather than a true analytical Delta, but its simulation error is sufficiently small for the comparisons conducted.

Each notebook was designed to follow a similar general structure, with option-specific modifications where required. In particular, the structure of the notebooks is as follows:

1. **General setup:** definition of the parameters, computation of reference Deltas and preliminary implementation checks.
2. **Finite differences:** evaluation of the forward and central methods on an N - h MSE grid, approximate selection of optimal h values, and assessment of CRN.
3. **Pathwise differentiation:** evaluation on the moneyness- σ grid, followed by any option-specific experiments or adaptations.
4. **Likelihood-ratio:** evaluation on the moneyness- σ grid, followed by an assessment of payoff centring.
5. **Comparative analysis:** comparison of the best versions of the methods using MSE and Runtime $\times$ MSE across the moneyness- σ grid.

Notice that a two-dimensional moneyness- σ grid was used to test error values under these varying parameters, while other parameters were generally fixed. Moneyness and volatility were selected because they are major determinants of option-payoff behaviour and estimator performance. The grid covers out-of-the-money, at-the-money and in-the-money options across both low- and high-volatility regions, while keeping the computational cost of the experiments reasonable. A summary of the principal parameters investigated and their fixed or variable values is given in Table 3.

Table 3: Summary of the principal experimental parameters.

Parameter	Values	Purpose
K, T, r	$K = 100, T = 1, r = 0.05$	Common fixed parameters
S_0/K	25 values in $[0.7, 1.3]$	Moneyness grid
σ	Baseline 0.25; 25 values in $[0.10, 0.60]$	Volatility grid
N	10^3 – 10^6 for tuning; 10^3 for comparisons	Simulation budget
h	Option- and N -dependent grids	Finite-difference tuning
M	Asian: 4 baseline (1 – 10^3 tested); barrier: 10	Averaging or simulation intervals
Q	10	Digital-call payment
B	$\{101, 110, 150, 250\}$	Barrier proximity
ε	25 values in $[10^{-5}, 10^2]$	Digital pathwise smoothing
c^*	Independent pilot sample of 10^3 paths	LR payoff centring

In addition to this, note that MSE was only investigated on an N - h grid for finite-difference methods, while their moneyness- σ analysis was reserved for the final comparative section. This decision was made in order to obtain approximately optimal choices of h as functions of N and, where relevant, other option-specific parameters. It also avoided an excessive number of experiments or grids for these methods prior to the comparative analysis. In their standard forms, pathwise differentiation and the likelihood-ratio method do not require a direct analogue of h , so their individual analyses already used the moneyness- σ grid.

Furthermore, some option- or method-specific experiments were conducted in the relevant notebooks. The computation of an approximately optimal smoothing parameter ε for the pathwise digital-call estimator and the estimation of the optimal centring constant c^* for the likelihood-ratio method are examples of this. These parameters were selected in the preceding estimator-specific experiments before being used in the final comparisons.

The final comparative sections evaluate central differences with CRN, the appropriate naive or adapted pathwise estimator, and the centred likelihood-ratio estimator using the same number of simulations, parameter grid and number of repeated trials. The computational cost of estimating c^* was included in Runtime $\times$ MSE, except in cases where the likelihood-ratio method was already suboptimal. In those cases, a best-case likelihood-ratio runtime excluding the computation of c^* was

considered, so that its performance was assessed under a convention favourable to it.

Finally, results were drawn both numerically and visually. The main conclusions stem from the computed error and runtime values, while plots were used to identify systematic patterns across the parameter grids and regional differences in estimator performance that would be difficult to recognise from numerical tables alone.

5 Results

The results presented below appear in the order in which they were obtained. Note that not every finding discussed in the `.ipynb` files is presented here; this section outlines the most relevant and distinctive results found and avoids repetition of similar results obtained across different options. The full tables of numerical values, together with further analysis and additional plots, can be found in the notebooks. Since Monte Carlo results may vary slightly between independent notebook runs, small numerical differences should be expected. The values presented below should therefore be interpreted as empirical estimates, while the conclusions are based primarily on stable patterns and orders of magnitude.

5.1 European Call

5.1.1 Finite Differences

To begin with, optimal values of h as a function of N were found in order to improve the Forward- and Central-Difference estimators and find an optimal balance between minimising estimator bias and variance. Under the fixed parameter values specified in Table 3, the empirical MSE of the approximations was computed on an N - h grid, using independent random numbers for the perturbed price estimates. This allowed the coefficients of the theoretically predicted MSE functions to be fitted to the two-dimensional data. As an example, Figure 1 below presents the MSE surface obtained for the Central-Difference method.

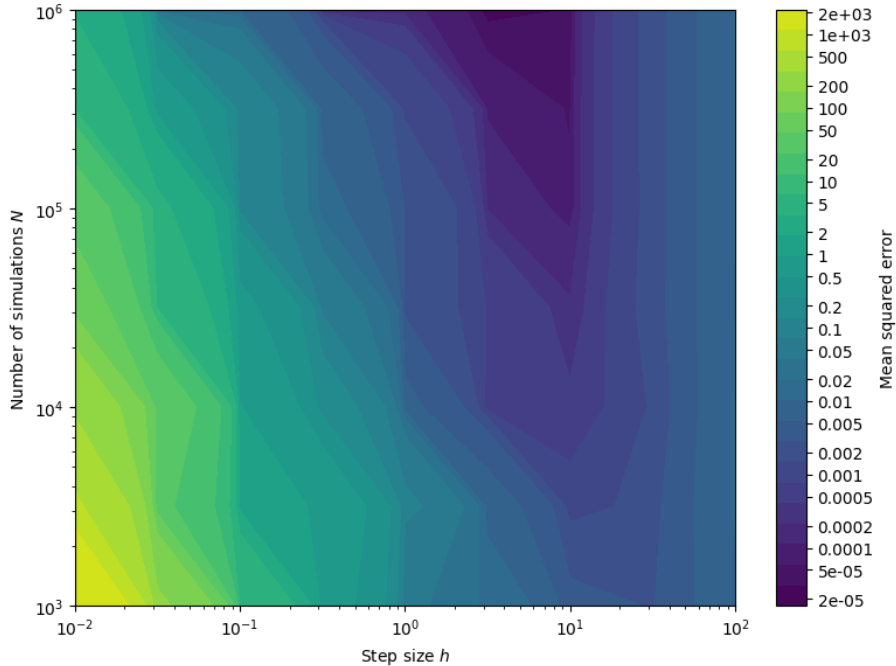


Figure 1: Empirical MSE of the Central-Difference estimator across the N - h grid under the fixed European call parameters. Independent random numbers are used for the two perturbed price estimates.

Following its theoretical form, the fitted Central-Difference MSE function was

$$\widehat{\text{MSE}}_{\text{CD}}(N, h) = (5.214590 \times 10^{-8}) h^4 + \frac{1.730521 \times 10^2}{Nh^2}.$$

The corresponding optimal- h function was therefore

$$\hat{h}_{\text{opt,CD}}(N) = \left(\frac{1.730521 \times 10^2}{2N(5.214590 \times 10^{-8})} \right)^{1/6} = 34.4076N^{-1/6}.$$

This function was used in the comparative section presented below to estimate appropriate step-size choices. In addition, Common Random Numbers (CRN) were implemented with the aim of lowering estimator variance. For a comparison of the results with and without CRN, the MSE at different values of h , under $N = 1,000$, is presented for the standard and CRN versions of the Central-Difference estimator in Table 4.

Table 4: Empirical MSE of the Central-Difference estimator with and without CRN for $N = 1,000$.

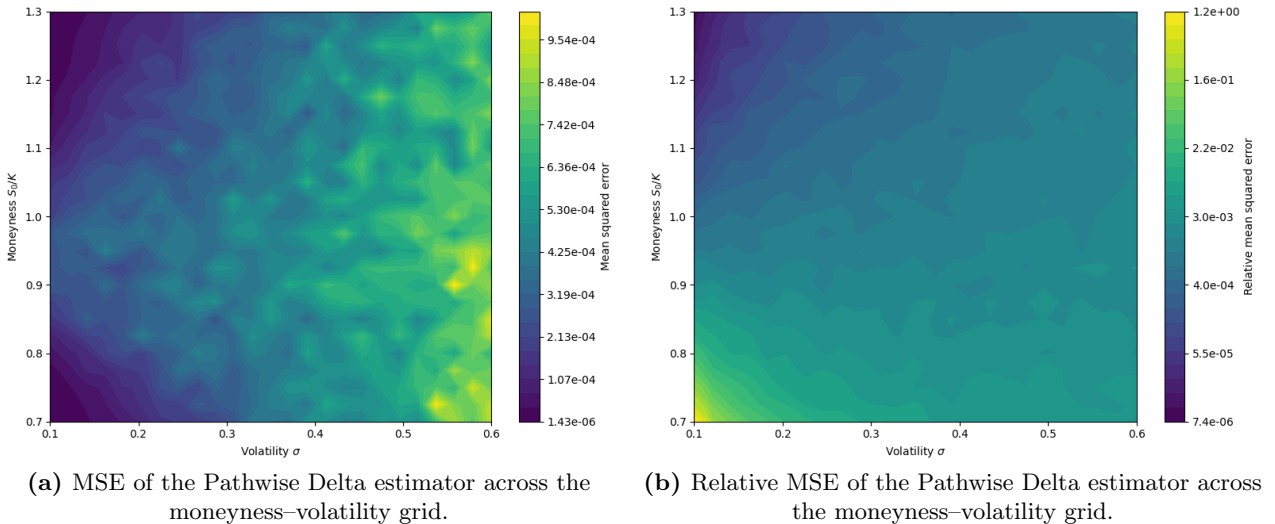
	$h = 0.01$	$h = 0.1$	$h = 1$	$h = 3.162278$
Independent shocks	3.0400×10^3	4.8370	2.1086×10^{-1}	1.2939×10^{-2}
With CRN	2.2604×10^{-4}	5.1479×10^{-5}	7.2500×10^{-4}	2.1520×10^{-3}

Clearly, CRN reduce the MSE drastically for the European call under the parameter values considered, particularly when h is small. Since the MSE was largely stabilised across smaller values of h once CRN were introduced, no separate optimal- h functions were derived for these estimators.

5.1.2 Pathwise Differentiation

Here, the MSE results on the moneyness–volatility grid are directly presented. For European calls, the Pathwise method does not require any modifications and does not present implementation nuances. Figure 2a shows that MSE is generally largest for ATM options and increases with volatility. Around the money, the exercise indicator is most uncertain, as simulated paths frequently end on either side of K , increasing estimator variance. Additionally, Figure 2b shows the relative MSE of the estimations. The ratios are smaller for deep ITM options, as their Deltas are larger.

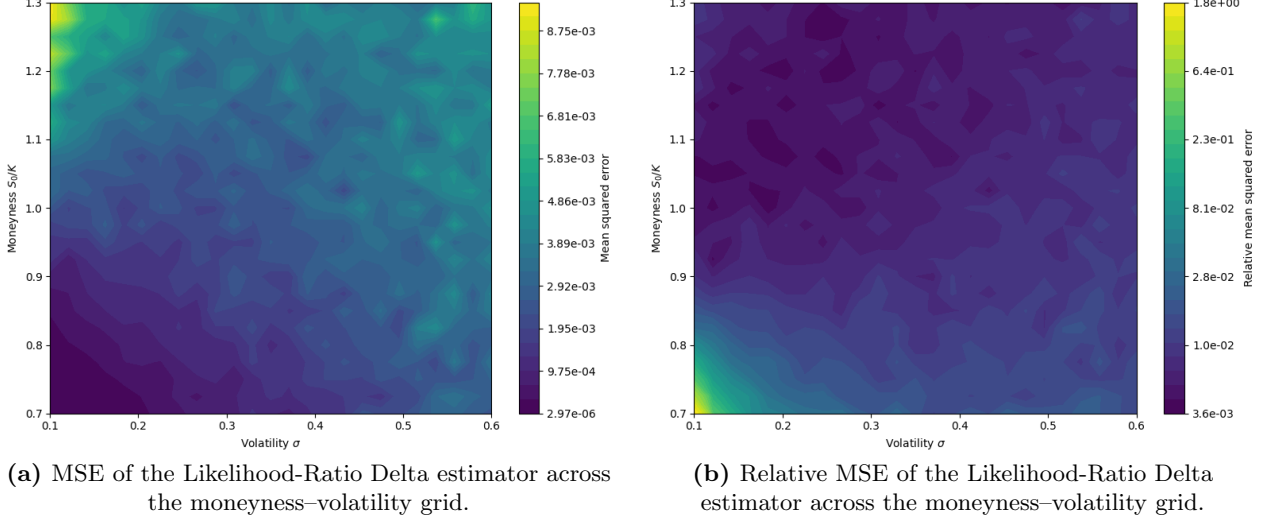
Figure 2: Pathwise Delta estimation errors.



5.1.3 Likelihood-Ratio

Similar qualitative results were obtained for the relative MSE of the Likelihood-Ratio method. However, the absolute MSE for deep ITM options shows a pattern opposite to the Pathwise case. This time, low-volatility options exhibit a high MSE; this is due to the score $Z/(S_0\sigma\sqrt{T})$ in the estimator, which results in a variance growing proportionally to $1/\sigma^2$. Figures 3a and 3b below exhibit these results.

Figure 3: Likelihood-Ratio Delta estimation errors.



In addition, Payoff Centring was found to be highly effective in reducing the MSE of the estimations, especially for the problematic high-moneyness, low-volatility region. Figure 4 below shows the improvements by considering the ratio $\text{MSE}_{\text{centred LR}} / \text{MSE}_{\text{ordinary LR}}$, for which values below 1 represent an improvement.

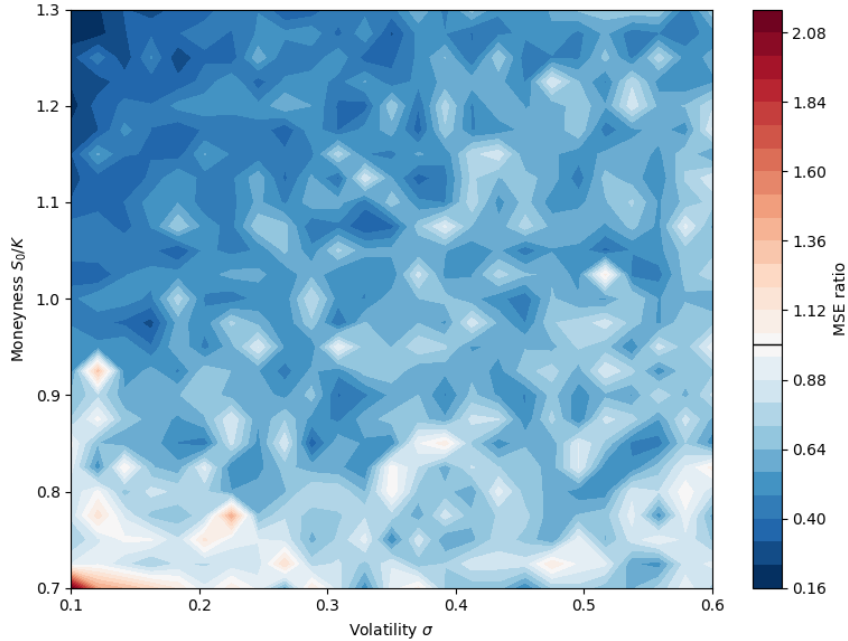


Figure 4: Ratio of the MSE of the centred Likelihood-Ratio estimator to that of the ordinary Likelihood-Ratio estimator. Values below 1 indicate that Payoff Centring reduces MSE.

5.1.4 General Comparison

To finish the investigation of the suitability of the different estimators for the European call Delta, we present the $\text{Runtime} \times \text{MSE}$ values under each estimator. Refer to Section 4 for details on the parameters tested and the variant of each method included.

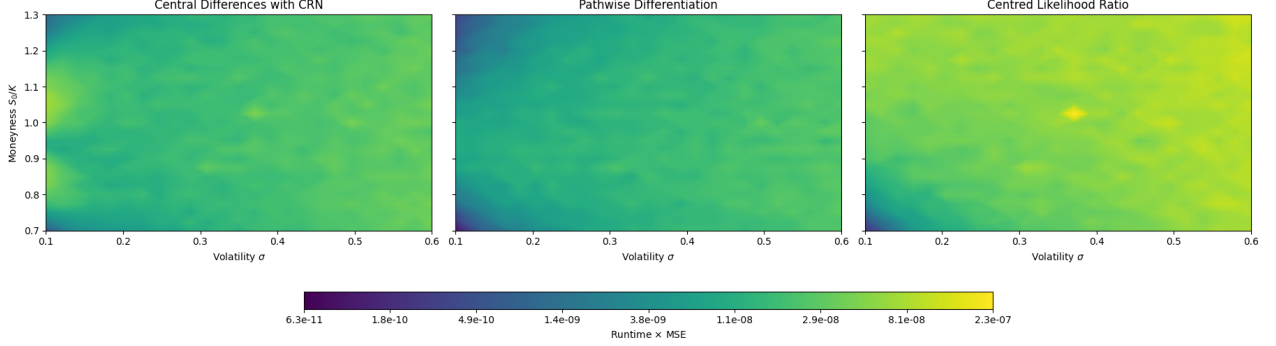


Figure 5: Runtime $\times$ MSE across the moneyness–volatility grid for Central Differences with CRN, Pathwise Differentiation and the centred Likelihood-Ratio method.

Clearly, Pathwise Differentiation is superior by this metric and dominates across the entire moneyness–volatility grid. This shows that its greater accuracy is not offset by a higher computational cost. On the other hand, the Likelihood-Ratio method is noticeably inferior to the rest, except for the low-moneyness, low-volatility region, where all methods agree closely. As explained in Section 4, the cost of computing c^* was excluded here, so the comparison already represents a favourable runtime convention for the Likelihood-Ratio method.

5.2 Asian Call

5.2.1 Finite Differences

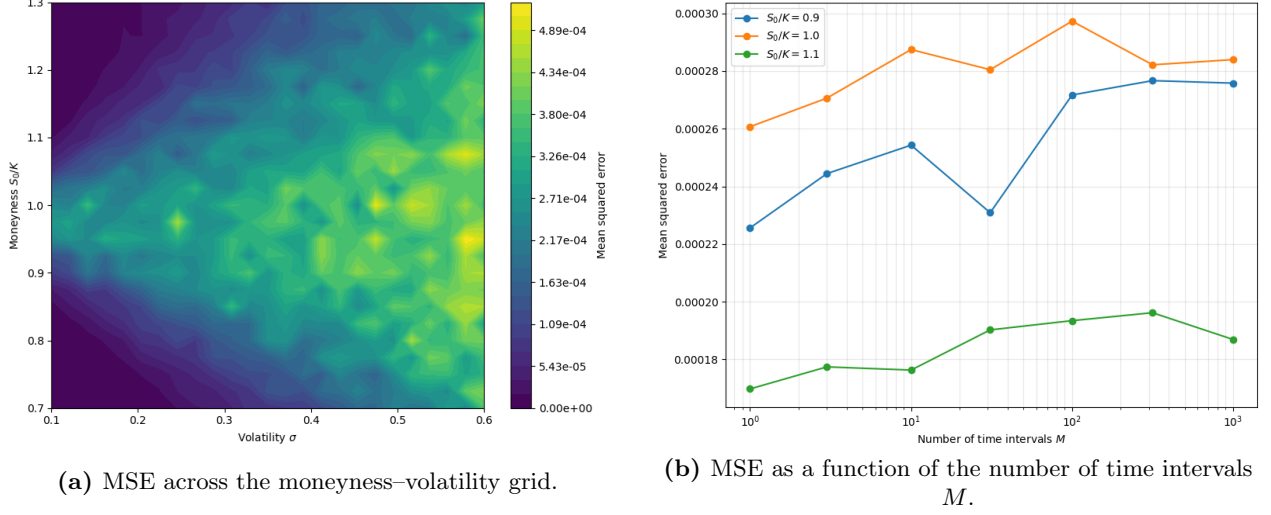
Both the Forward- and Central-Difference methods exhibited results for Asian calls that were very similar to those obtained for European calls. The Central-Difference method exhibited a lower MSE across the investigated $N-h$ grid due to its lower bias, while the use of CRN once again reduced the MSE of the estimations. Although the MSE values were lower overall, path dependence changed the numerical magnitude of the errors but not the fundamental finite-difference bias–variance behaviour observed for European calls. The complete $N-h$ results are provided in `notebooks/03_asian_call.ipynb`.

5.2.2 Pathwise Differentiation

The Pathwise estimates again showed very similar behaviour to the European call case. In particular, the same “funnel” was observed in Figure 6a, where ATM options have higher MSE values and MSE generally increases with volatility. This is explained by the same reasons highlighted in the European call investigation.

A more interesting experiment, however, involved varying the number of time intervals M under fixed remaining parameters to observe the resulting changes in MSE. As seen in Figure 6b, M does not appear to affect the MSE of the Pathwise estimations in a clear or systematic manner. This suggests that the Pathwise estimator may be a good candidate when the MSE needs to remain comparatively stable as the number of monitoring dates varies. In addition, the different moneyness values again show that MSE is greatest for ATM options, since their simulated arithmetic averages are the most likely to lie near the exercise boundary $A = K$.

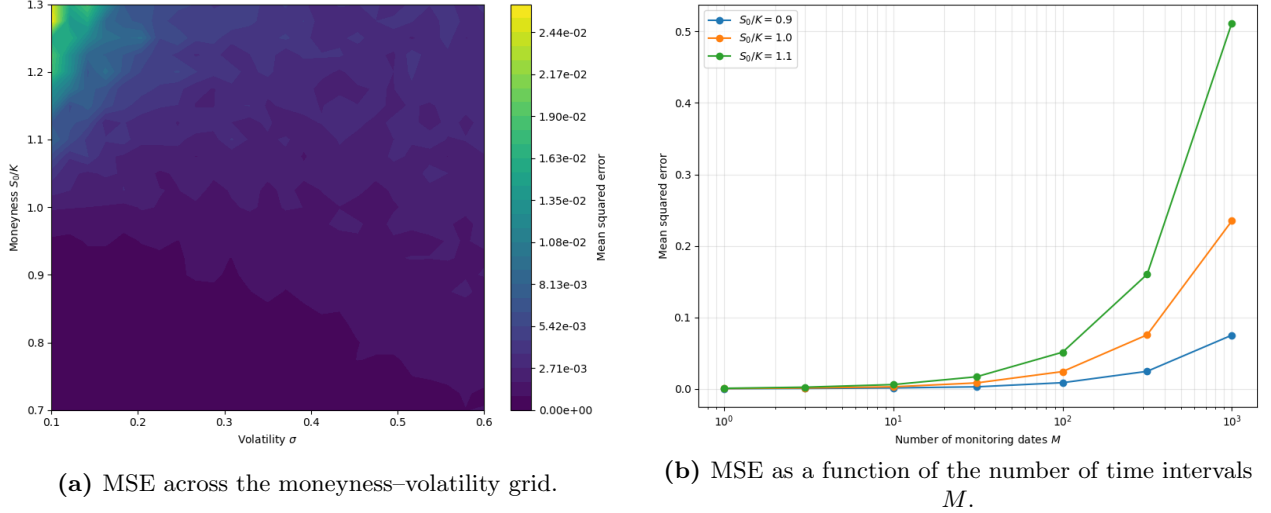
Figure 6: MSE behaviour of the Pathwise Asian call Delta estimator.



5.2.3 Likelihood-Ratio

Likelihood-Ratio estimates exhibited similar behaviour to their European equivalents. The high-moneyness, low-volatility region presented the highest MSE due to the $1/\sigma^2$ factor in the variance and the fact that the payoff was non-zero for almost every simulated path. Figure 7a shows this behaviour. Unlike for Pathwise estimates, monitoring frequency has a clear effect on the MSE, which again stems from the dependence of the likelihood-ratio weights on M . Figure 7b exhibits this dependence for different moneyness values.

Figure 7: MSE behaviour of the Likelihood-Ratio Asian call Delta estimator.



A separate log-log fit presented in `notebooks/03_asian_call.ipynb` produced the approximate relationship

$$\text{MSE} \propto M^{0.83}$$

for the ATM case. The exponent is approximate and may vary slightly between independent notebook runs. This near-linear growth is partly caused by the explicit dependence of the score

$$W_M = \frac{Z_1}{S_0 \sigma \sqrt{T/M}}$$

on $\sqrt{M}$, and hence the dependence of its square on M . However, the Asian payoff H_M and its joint distribution with Z_1 also depend implicitly on M , so the observed exponent is not determined by the score alone.

The effect of Payoff Centring on the MSE was generally similar to that observed for European calls. An exception occurs in the extreme low-moneyness, low-volatility region, where the ordinary LR variance is already close to zero and centring may instead worsen the result. Further details may be found in `notebooks/03_asian_call.ipynb`.

5.2.4 General Comparison

Comparative Runtime $\times$ MSE plots for all methods on the usual moneyness–volatility grid were computed for different numbers of time intervals: $M = 4$, $M = 252$ and $M = 2016$, representing quarterly, daily and hourly monitoring, respectively. Figure 8 presents the results for quarterly monitoring.

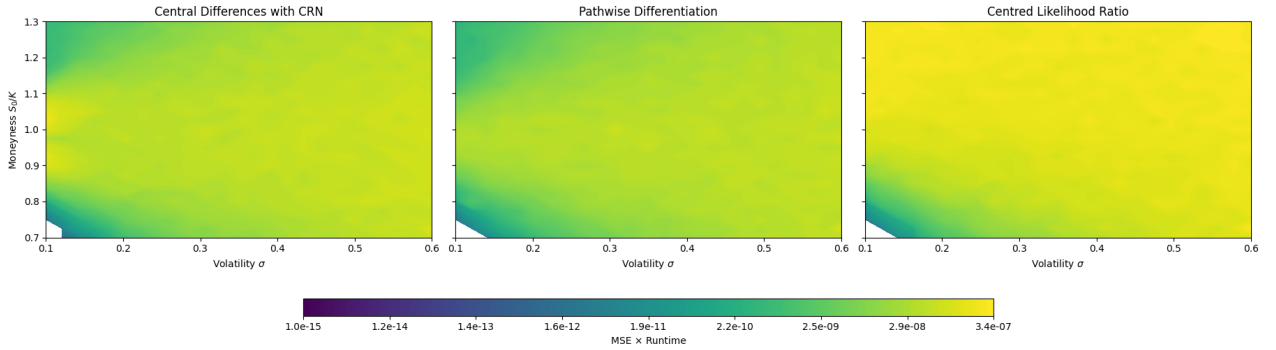


Figure 8: Runtime $\times$ MSE across the moneyness–volatility grid for Central Differences with CRN, Pathwise Differentiation and the centred Likelihood-Ratio method, using quarterly monitoring ($M = 4$). A common logarithmic colour scale is used across the three panels.

Notice that the white regions on the plots correspond to empirical MSE values of zero. These occur because no simulated paths activate the payoff in some deeply OTM, low-volatility cases, and therefore cannot be represented on the logarithmic colour scale.

In general, we observe dominance by the Pathwise Differentiation method, with similar behaviour from the Central-Difference method with CRN. This similarity illustrates the known theoretical result presented in `notebooks/01_estimator_theory.ipynb`: under CRN, Finite-Difference methods approach Pathwise behaviour as $h \rightarrow 0$. Increasing N allows the approximately optimal value of h to decrease while keeping the resulting sampling variance under control.

On the other hand, the Likelihood-Ratio method clearly exhibits worse error behaviour across practically the entire grid. Moreover, Figure 8 considers only $M = 4$, the lowest monitoring frequency investigated. Since LR error increases approximately as a power of M , while the Pathwise MSE remains relatively constant as M increases, the disadvantage of LR becomes more pronounced at higher monitoring frequencies. The daily and hourly comparisons presented in `notebooks/03_asian_call.ipynb` support the same conclusion. Thus, for Asian calls, Delta is better estimated through the method of Pathwise Differentiation.

5.3 Digital Call

5.3.1 Finite Differences

Both the Forward- and Central-Difference methods exhibited N - h trade-offs very similar to the European and Asian cases. This is, in general, due to the fact that even though the digital call payoff function is discontinuous, its option price function remains smooth under the BSM model, and finite differences approximate the derivative of the price rather than that of the payoff. Optimal- h functions were also derived for both methods, and CRN were implemented. The best finite-difference variant was found to be the Central-Difference estimator under an optimal choice of h and CRN, which is the variant used in the later comparative analysis.

5.3.2 Pathwise Differentiation

Since Pathwise Differentiation cannot be applied directly to the discontinuous digital payoff, a naïve implementation of the method is not possible. In fact, the pathwise derivative of the digital indicator is zero almost surely, so the naïve estimator misses the Dirac-like contribution concentrated at $S_T = K$. As detailed in `notebooks/01_estimator_theory.ipynb`, the step function that defines the digital call payoff was approximated using normal CDFs with varying standard deviations (Liu and Hong, 2011). This required the choice of a smoothing parameter ε , which was optimised through the same surface-fitting technique used to estimate optimal values of h . For fixed moneyness $S_0/K = 1$ and volatility $\sigma = 0.25$, Figure 9 shows the MSE as a function of ε and N .

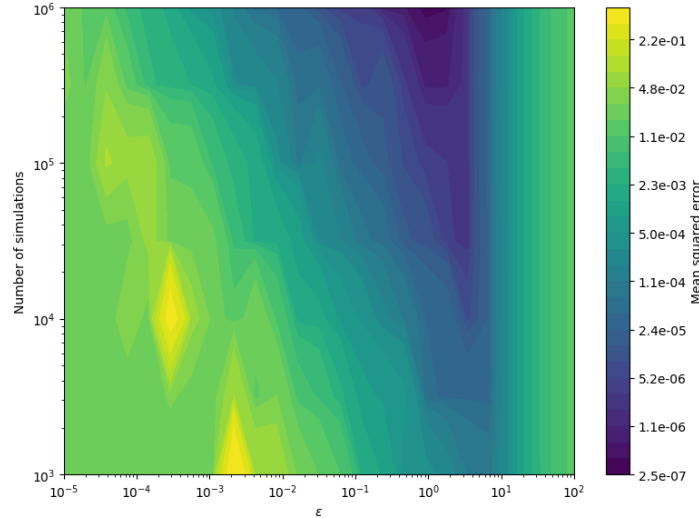


Figure 9: MSE of the smoothed Pathwise digital call Delta estimator as a function of the smoothing parameter ε and the number of simulations N .

This allowed for the derivation of an optimal- ε function satisfying

$$\varepsilon_{\text{opt}}(N) \propto N^{-1/5}.$$

Using this function, MSE was investigated on the usual moneyness–volatility grid for varying optimal (N, ε) pairs. Figure 10 presents the two endpoint cases, while the intermediate results are provided in `notebooks/04_digital_call.ipynb`.

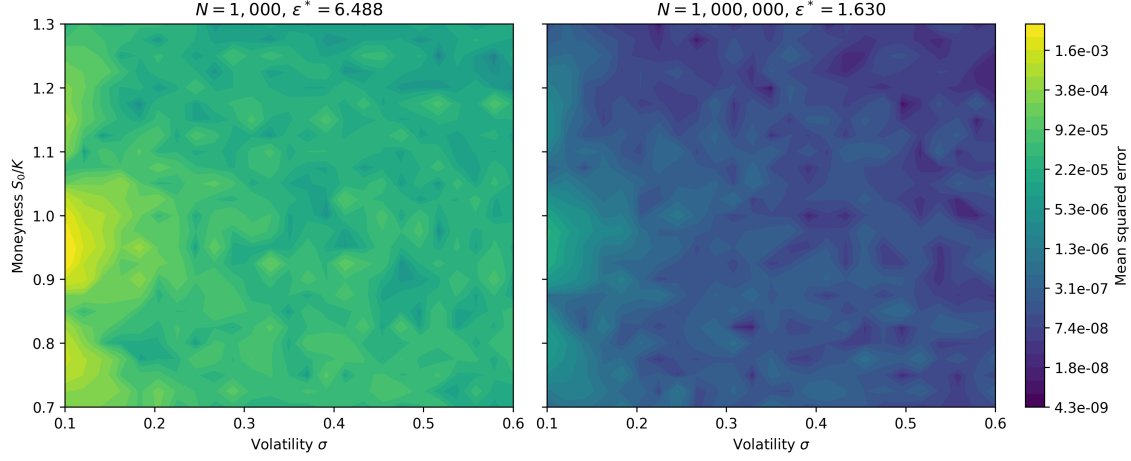


Figure 10: MSE of the smoothed Pathwise digital call Delta estimator across the moneyness–volatility grid for $N = 1,000$ and $N = 1,000,000$, using the corresponding optimal values of ε . A common logarithmic colour scale is used across both panels.

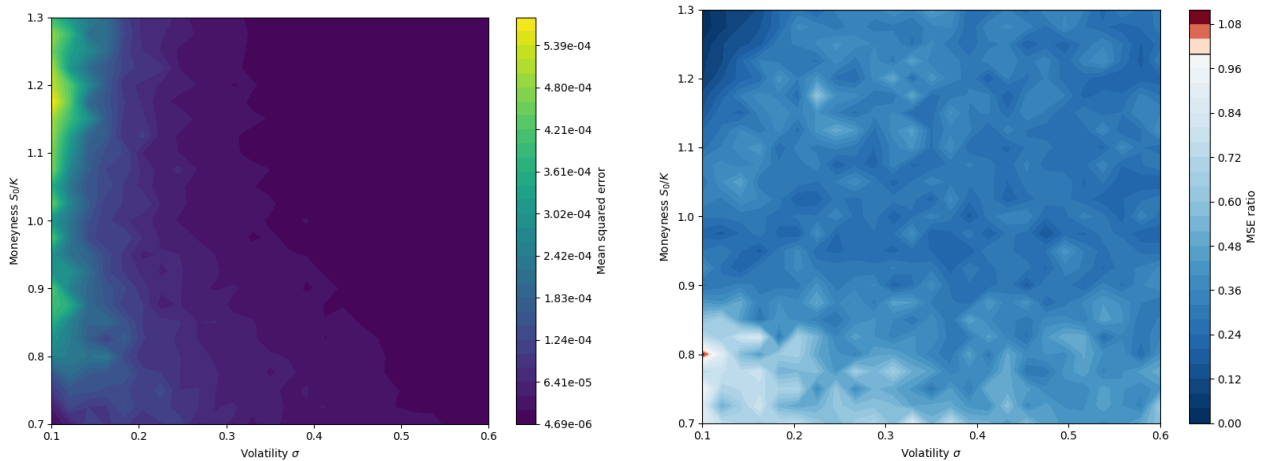
Figure 10 shows that MSE is greatest for low-volatility, near-ATM options. This is explained by the fact that Delta is largest near the exercise boundary between zero and a fixed positive payoff. In addition, low volatility concentrates the terminal asset price around this boundary for near-ATM options, increasing the variance of the smoothed estimator. At both moneyness extremes, by contrast, Delta approaches zero.

5.3.3 Likelihood-Ratio

Unlike the Pathwise estimator, the Likelihood-Ratio estimator handles discontinuities of payoff functions because it does not attempt to differentiate them. Details on this advantage can be found in `notebooks/01_estimator_theory.ipynb`. Consequently, the behaviour of this estimator is not very different from that of its European and Asian equivalents. The primary driver of MSE is σ , with smaller values producing larger errors because of the inverse dependence of the score on volatility. Nonetheless, moneyness appears to have little to no effect on MSE, as Figure 11a shows.

As usual, Payoff Centring was implemented, resulting in significant MSE improvements across most of the grid. Figure 11b highlights these improvements, although a small number of extreme low-moneyness, low-volatility cases show little or no benefit.

Figure 11: MSE behaviour of the Likelihood-Ratio digital call Delta estimator.



(a) MSE across the moneyness–volatility grid.

(b) Effect of Payoff Centring on MSE.

5.3.4 General Comparison

Since the effect of increasing N was already investigated within each method and MSE is proportional to Q^2 , both variables were fixed in the comparative analysis. The two usual comparative metrics were then computed across a single moneyness–volatility grid, fixing all remaining variables and using the best variant of each estimator. Figure 12 presents the Runtime $\times$ MSE results, while the corresponding absolute MSE comparison is provided in `notebooks/04_digital_call.ipynb`.

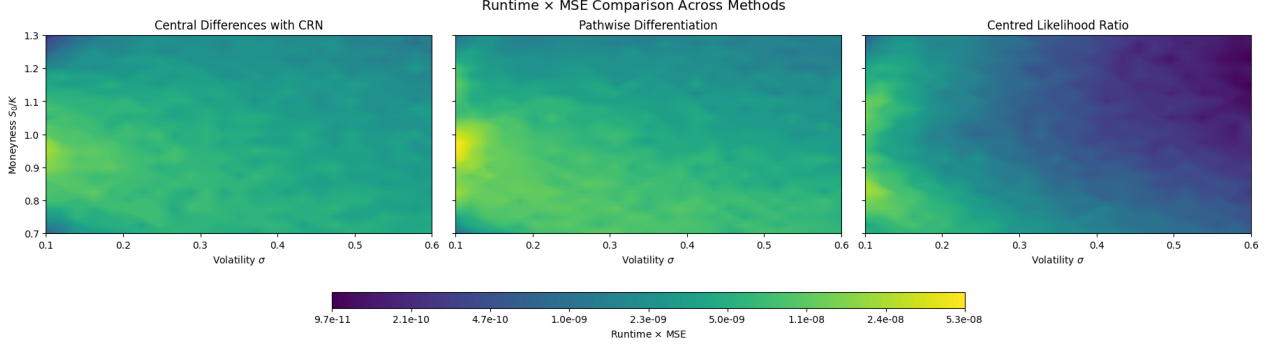


Figure 12: Runtime $\times$ MSE across the moneyness–volatility grid for Central Differences with CRN, smoothed Pathwise Differentiation and the centred Likelihood-Ratio method. The runtime required to estimate the centring constant c^* is included, and a common logarithmic colour scale is used across the three panels.

We observe that the Likelihood-Ratio method with Payoff Centring dominates the other two methods across practically the entire grid, even after accounting for the runtime required to compute the centring constant c^* . This dominance is especially clear for high-volatility options. On the other hand, under the Runtime $\times$ MSE metric, the adjusted method of Pathwise Differentiation presents the worst results, even after choosing an approximately optimal value of ε . Although the Central-Difference method with CRN generally presents worse absolute MSE results than smoothed Pathwise Differentiation, its lower runtime makes it preferable under the combined metric.

These results suggest that, for options with discontinuous payoff functions, the Likelihood-Ratio method with Payoff Centring may be better suited than modified Pathwise Differentiation methods for estimating Delta. This conclusion appears to hold across a wide range of moneyness and volatility values.

5.4 Barrier Call

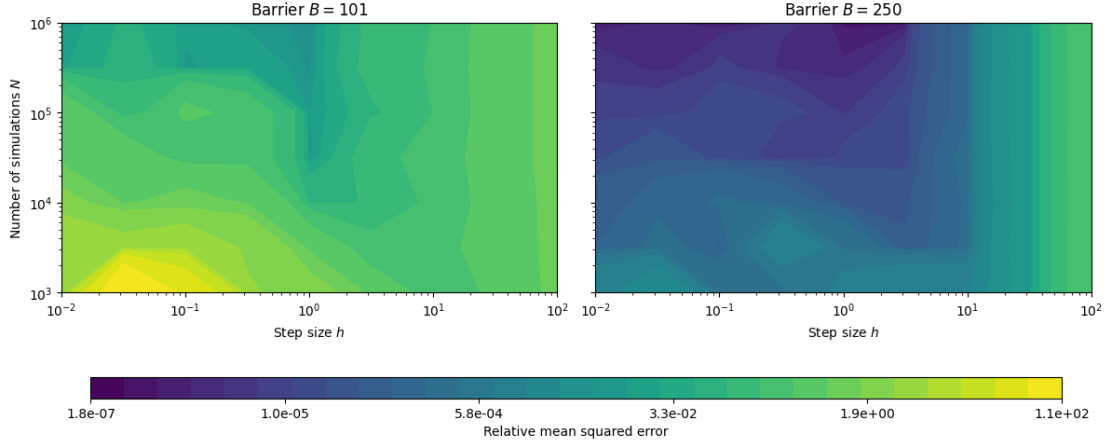
In this section, one of the primary effects investigated is that of barrier proximity on estimator MSE. Since MSE can vary substantially across different barrier values, the entire section focuses on relative rather than absolute MSE values. Additionally, recall that all methods use Brownian-bridge conditioning to account for potential barrier crossings of the continuously monitored barrier call that may remain undetected between consecutive simulated values (Glasserman, 2004; Glasserman and Staum, 2001).

5.4.1 Finite Differences

In order to draw better comparisons between estimators later in this section, the MSE of the Forward and Central Difference estimators was computed on an N – h grid. This time, in order to avoid the complexity of fitting an MSE surface as a function of N and h for this specific option, whose behaviour also varies with the barrier, the MSE-minimising step sizes were estimated numerically from the tables of values obtained. Figure 13 shows the Central Difference results for the two most

extreme barrier values investigated.

Figure 13: Relative MSE of the Central Difference estimator with CRN on the N - h grid for $B = 101$ and $B = 250$. Both panels use a common logarithmic colour scale.



Notice that step sizes for which the perturbed price approximations start in knockout states, such as $h = 10$ when $B = 101$, do not represent a local price-gradient approximation. The continuously monitored up-and-out call price is continuous at the barrier: as S_0 approaches B from below, the probability of surviving without touching the barrier tends to zero, and so does the option price. However, the price is generally not smooth across the knockout boundary. Therefore, a finite-difference stencil satisfying $S_0 + h \geq B$ crosses this boundary and produces a secant slope involving an already-knocked-out contract rather than a local approximation of Delta.

This helps explain why smaller values of h , which remain within the live region, generally provide more meaningful approximations for nearby barriers. An exception is the case $B = 250$, where even relatively large step sizes, such as $h > 10$, remain far from the knockout boundary and may still produce good Delta approximations.

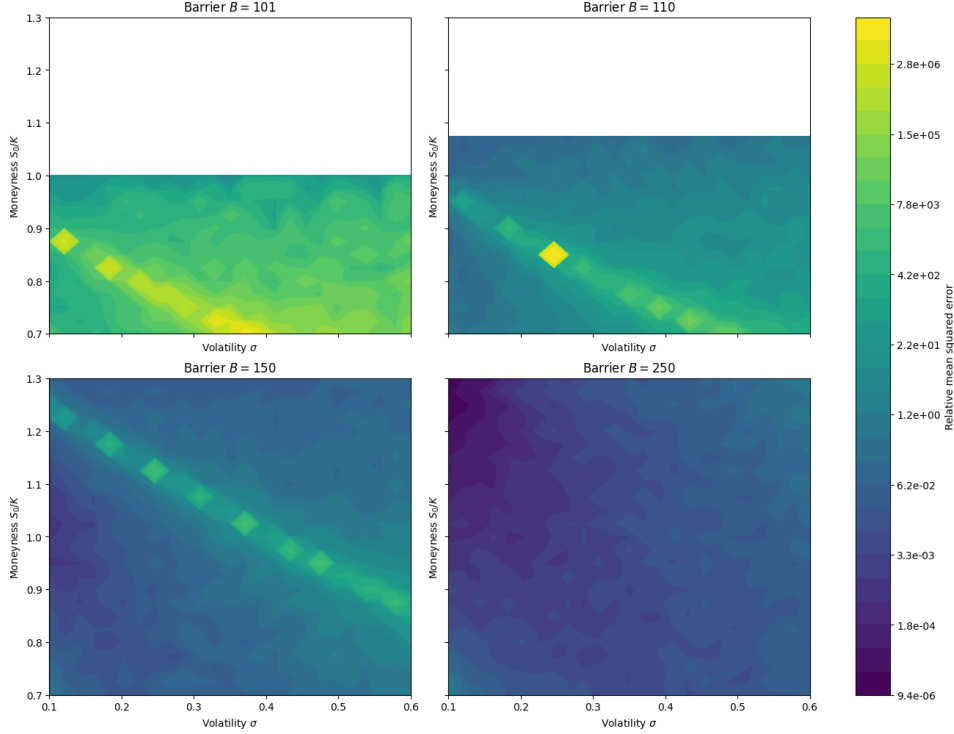
5.4.2 Pathwise Differentiation

The results obtained for the Pathwise method highlight a clear dependence of the MSE on the barrier value. As B becomes larger, the MSE becomes significantly lower across the entire moneyness-volatility grid. This is explained by the fact that, as the barrier moves away from the initial underlying asset price and the strike price, the option behaves increasingly similarly to a European call, since the event of crossing the barrier during the life of the option becomes increasingly rare. Thus, one should expect the MSE of the Pathwise estimates to approach that of the European call Pathwise estimates as the barrier increases. Figure 14 exhibits this effect.

Notice that the white regions correspond to undefined MSE values. These occur when $S_0 \geq B$, meaning that the option is knocked out from the start of the contract and its relative MSE is therefore undefined.

Additionally, a high-MSE diagonal ridge can be observed across the plots. This is caused by the underlying reference Delta at those points of the grid being very close to zero, which results in high relative errors even when the absolute estimation error is not particularly large. In this diagonal region, two opposing effects approximately balance each other: increasing S_0 raises the probability and size of an in-the-money terminal payoff, but it also raises the probability of crossing the barrier and knocking out. For $B = 250$, the barrier is sufficiently high that no moneyness-volatility combination in the grid produces this effect; there, the effect of finishing above the strike price dominates because crossing the barrier is rare. Lastly, notice that the ridge follows this diagonal trend because high-moneyness options require less volatility for the knockout effect to become significant, while

Figure 14: Relative MSE of the Brownian-bridge-adapted Pathwise estimator across the moneyness-volatility grid for $B = 101$, $B = 110$, $B = 150$ and $B = 250$. The common logarithmic colour scale shows the movement and eventual disappearance of the diagonal high-error ridge.



lower-moneyness options require higher volatility for the same effect.

5.4.3 Likelihood-Ratio

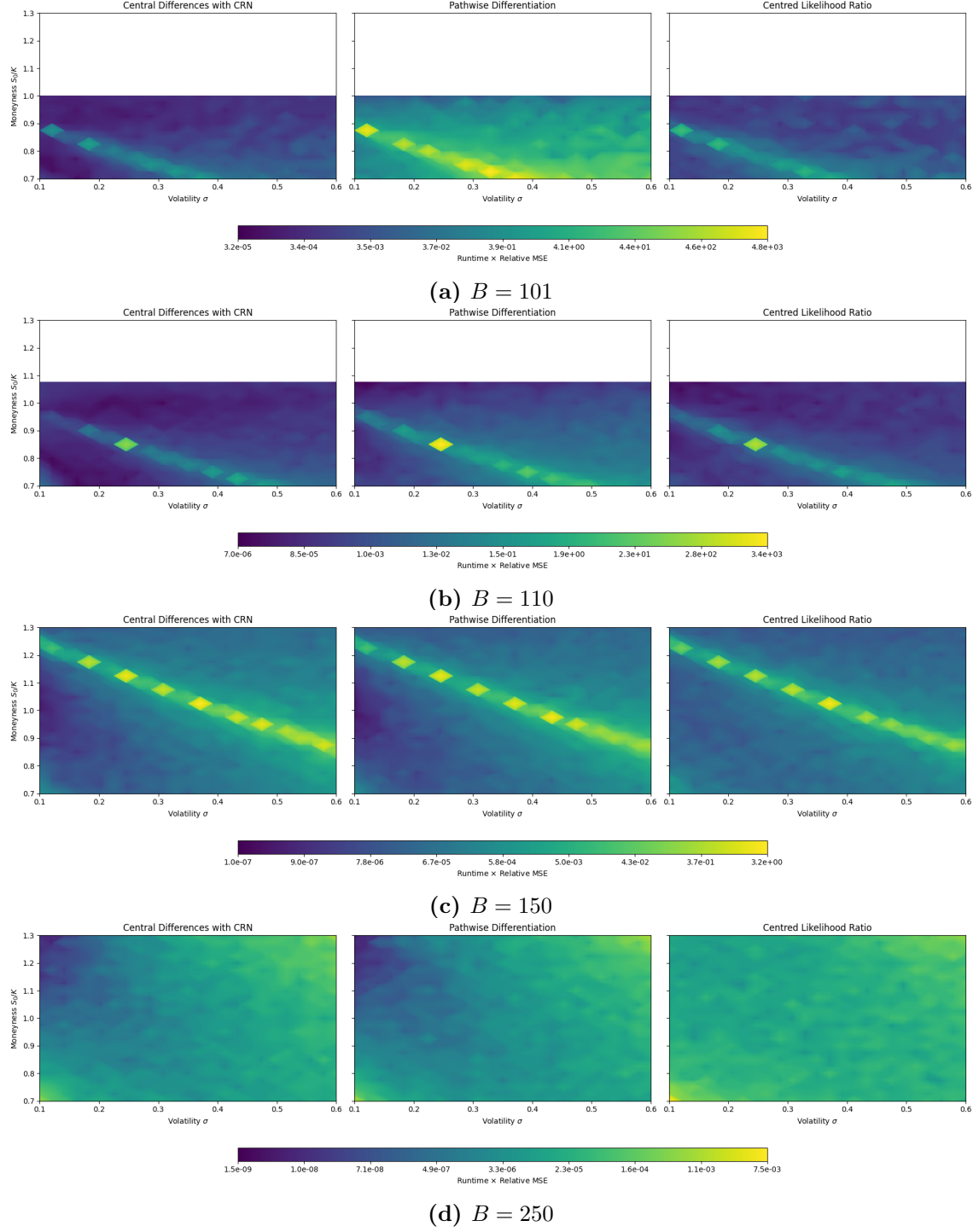
A very similar pattern to that of the Pathwise method was observed for the Likelihood-Ratio method, allowing the same qualitative conclusions to be drawn. In particular, the diagonal region of high relative MSE moves as the barrier increases and disappears once the barrier becomes sufficiently remote. To avoid repeating the Pathwise results, the corresponding plots are omitted here and may be found in `notebooks/05_barrier_call.ipynb`. The differences between the two estimators are instead considered in the comparative section below.

It should be noted that, on this occasion, Payoff Centring was not found to provide substantial and systematic improvements in the MSE of the estimations for low-barrier options. For high-barrier options, its effect became closer to that observed for European calls, with the clearest improvements occurring in the high-moneyness, low-volatility region and more moderate improvements elsewhere. Further details and plots may be found in the notebook.

5.4.4 General Comparison

On this occasion, both the relative MSE and $\text{Runtime} \times \text{relative MSE}$ comparisons lead to the same general conclusions. In particular, the dominant method depends on the proximity of the barrier. Figure 15 presents the $\text{Runtime} \times \text{relative MSE}$ results, while the corresponding relative MSE plots may be found in the notebook.

Figure 15: Runtime $\times$ relative MSE across the three estimators for different barrier values. A common logarithmic colour scale is used across the three methods within each barrier value. The Likelihood-Ratio runtime includes the computation of the centring constant c^* .



For low-barrier values, the Central Difference method under the numerically selected value of h and CRN produces the lowest values over most of the valid grid. The Likelihood-Ratio method with Payoff Centring exhibits similar but generally worse results. On the other hand, the Delta of high-barrier options is better approximated through Pathwise Differentiation over most of the grid, with the Central Difference method showing very similar results. This is consistent with the previous findings, since high-barrier options behave similarly to European options, for which Pathwise estimates were found to be superior.

It should be noted, however, that the empirical choice of h was made solely according to the lowest MSE in the tables, regardless of whether it resulted in a local derivative approximation. In particular, the selected value $h = 10$ crosses or reaches the knockout boundary at many points for $B = 101$ and $B = 110$. Under the unrestricted numerical optimisation used here, Central Differences with CRN are therefore the best-performing method for low barriers in this particular comparison. However, this apparent dominance is likely partly coincidental and should not be regarded as a reliable general result. If h is restricted so that the finite-difference stencil remains within the live region and represents a local derivative approximation, the Likelihood-Ratio method provides better results over most of the low-barrier grid. Further results under this restriction may be found in the notebook.

It is thus clear that, when the barrier is sufficiently remote that the contract behaves approximately like a European call and its discontinuity is rarely activated, Pathwise Differentiation dominates. Conversely, when the barrier is close and the payoff discontinuity has a significant effect, Delta may be better estimated through the Likelihood-Ratio method once finite-difference step sizes are restricted to valid local perturbations. Finite Difference methods, and in particular Central Differences with CRN, remain a reliable and competitive choice across different barrier values because of their robustness and general applicability.

6 Discussion

The results obtained across the different options considered highlight the main properties of each Delta estimator and allow one to evaluate their appropriateness depending on the principal features and regularity of the payoff functions.

Firstly, Finite-Difference methods were found to produce consistently reliable estimators across the different payoff structures investigated. In particular, the Central-Difference method with CRN and an appropriately selected step size h was found to be the best variant of these methods. For sufficiently regular payoffs, its behaviour remained close to that of Pathwise Differentiation, consistent with the convergence of Central Differences with CRN towards the corresponding Pathwise estimator as $h \rightarrow 0$ (Haugh, 2017). In discontinuous settings, it generally remained somewhat inferior to the Likelihood-Ratio method when the discontinuity had a significant effect, but continued to provide competitive estimates. Consequently, Central Differences with CRN were found to be generally reliable and robust, since their implementation did not require the payoff function to be replaced by an approximation. On the other hand, these estimators require an approximately optimal step size to be selected through an independent investigation, as an unsuitable value of h may produce unusually high bias or variance.

Additionally, the empirically MSE-minimising value of h may sometimes produce an approximation that is far from local, as seen for nearby barrier options. When the finite-difference stencil reaches or crosses the knockout boundary, the resulting estimate remains a genuine finite-difference quotient based on the true option-price function. However, it compares prices across a relatively wide interval containing the non-smooth knockout boundary rather than measuring the behaviour of the price function within the immediate neighbourhood of S_0 . Such a choice of h may happen to produce a low MSE for the particular parameters investigated, but this agreement may be specific to that configuration and should not be interpreted as evidence that similarly non-local step sizes will provide

reliably accurate Delta estimates more generally.

Secondly, Pathwise Differentiation was found to be the most accurate and computationally efficient method for the continuous, almost-surely pathwise-differentiable cases considered. In such cases, the method requires no parameter tuning and remains reliable under path dependence. For Asian calls, for example, its error remained comparatively stable as the number of averaging dates increased. This suggests that path dependence alone does not make Pathwise Differentiation unsuitable. However, the method requires sufficient pathwise differentiability rather than continuity alone. It cannot be applied naively to the discontinuous digital-call payoff, whose pathwise derivative is zero almost surely. Smoothing made the method applicable by replacing the original payoff with a differentiable approximation, but introduced bias, an additional tuning parameter and a corresponding bias–variance trade-off. Even after selecting an approximately optimal smoothing parameter, this estimator produced the worst results for the digital call under the $\text{Runtime} \times \text{MSE}$ metric.

Lastly, the Likelihood-Ratio method, and in particular its payoff-centred variant, was found to provide good Delta estimators for options with discontinuous payoffs. Since its definition differentiates the underlying probability density rather than the payoff function, it can accommodate payoff discontinuities without replacing the true payoff with a smooth approximation. Nonetheless, this method was highly sensitive to low volatility because of the inverse dependence of its score on σ . Its error also increased with the number of time intervals in the Asian-call investigation: under the fixed ATM parameters considered, the approximate empirical relationship $\text{MSE} \approx CM^{0.83}$ was obtained. Payoff centring produced substantial improvements across many of the investigated regions, but estimating the optimal centring constant c^* requires an additional pilot sample and therefore further computational effort. Although the true optimal value cannot increase the estimator variance relative to the uncentred choice $c = 0$, an inaccurate estimate of c^* may fail to provide the intended variance reduction and may instead worsen the empirical MSE. Overall, the Likelihood-Ratio method is better suited to cases in which the implementation of Pathwise Differentiation is prevented or complicated by payoff discontinuities.

In general, the results indicate that the best choice of Delta estimator depends primarily on payoff regularity and on whether any discontinuity has a material effect on the option. For European and Asian calls, Pathwise Differentiation dominates clearly, with Central Differences with CRN providing a close and more generally applicable alternative. For the digital call, the payoff-centred Likelihood-Ratio method is preferable. The barrier results present a transition between these cases. When the barrier is remote, the contract behaves increasingly like a European call and Pathwise Differentiation dominates. When the barrier is close and the payoff discontinuity becomes important, the Likelihood-Ratio method becomes preferable when the Central-Difference comparison is restricted to sufficiently local step sizes. Larger step sizes that reach or cross the knockout boundary occasionally produced lower empirical MSE values, but these correspond to comparatively non-local approximations whose numerical accuracy may be specific to the parameters considered. Central Differences with CRN nevertheless remained competitive across the different barrier values investigated. Among the methods and payoff structures investigated, therefore, Central Differences with CRN appear to provide the safest general-purpose baseline, despite rarely being the outright optimum.

It should be noted that the analysis conducted above presents several limitations. The entire investigation relies on BSM assumptions, which may not be suitable for modelling certain underlying assets. In addition, the analysis was limited to a fixed parameter grid, while several other variables that may affect estimator performance, such as the time to maturity T , were held fixed throughout. The computation of the Asian-call MSE also relied on a high-precision Pathwise estimate that itself contains simulation error relative to the unknown exact Delta. For barrier calls, relative MSE becomes unstable where the reference Delta is close to zero, while the smaller number of repeated trials makes fine regional comparisons less reliable. Finally, runtime measurements are dependent on the particular implementation and hardware used, so their precise magnitudes may vary across implementations.

7 Conclusion

This project aimed to study the properties of common Monte Carlo Delta estimators and to establish a general decision rule for choosing between them depending on the option's structure. This was done by considering the MSE and Runtime $\times$ MSE of the estimations. The results obtained and the analysis conducted highlight that, for sufficiently pathwise-differentiable payoff functions, the method of Pathwise Differentiation provides the best estimates. For options with discontinuous payoffs, however, the Likelihood-Ratio method generally provides more accurate estimators without requiring the payoff function to be smoothed or replaced by an approximation. Whenever the regularity of the payoff is unclear, or when the specialised implementation of these two methods is to be avoided, optimised Finite-Difference methods provide robust and reliable estimators.

Further investigation of these estimators could consider the effect of the time to maturity T , which is especially relevant for the Likelihood-Ratio method, as $T^{-1/2}$ appears as a multiplying factor in its score function. In addition, automatic tuning could be implemented into the estimator functions developed, which would allow approximately optimal parameter choices to be obtained without separate intermediate experiments. Such additions would further inform the general estimator decision rule and improve the practical usability of the Monte Carlo Delta estimation engine developed.

References

- Black, Fischer and Myron Scholes (1973). “The Pricing of Options and Corporate Liabilities”. In: *Journal of Political Economy* 81.3, pp. 637–654. DOI: 10.1086/260062.
- Broadie, Mark and Paul Glasserman (1996). “Estimating Security Price Derivatives Using Simulation”. In: *Management Science* 42.2, pp. 269–285. DOI: 10.1287/mnsc.42.2.269.
- Glasserman, Paul (2004). *Monte Carlo Methods in Financial Engineering*. New York: Springer. DOI: 10.1007/978-0-387-21617-1.
- Glasserman, Paul and Jeremy Staum (2001). “Conditioning on One-Step Survival for Barrier Option Simulations”. In: *Operations Research* 49.6, pp. 923–937. DOI: 10.1287/opre.49.6.923.10018.
- Haugh, Martin (2017). *Estimating the Greeks*. Lecture notes for IEOR E4703: Monte-Carlo Simulation. Columbia University. URL: https://www.columbia.edu/~mh2078/MonteCarlo/MCS_Greeks.pdf.
- Liu, Guangwu and L. Jeff Hong (2011). “Kernel Estimation of the Greeks for Options with Discontinuous Payoffs”. In: *Operations Research* 59.1, pp. 96–108. DOI: 10.1287/opre.1100.0844. URL: <https://doi.org/10.1287/opre.1100.0844>.
- Merton, Robert C. (1973). “Theory of Rational Option Pricing”. In: *The Bell Journal of Economics and Management Science* 4.1, pp. 141–183. DOI: 10.2307/3003143.